

Chapter 15: AI Governance

Learning Objectives

By the end of this chapter, you will:

- Understand the principles and importance of AI Governance in enterprise environments.
 - Learn how organizations establish policies, controls, and oversight for AI systems.
 - Develop frameworks for managing AI risks, compliance, and accountability as an AI TPM.
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Beginner Explanation

AI Governance is the set of rules, policies, and processes that ensure AI systems are:

- Safe
- Secure
- Reliable
- Compliant
- Ethical
- Accountable

Think of AI Governance as traffic rules for AI systems.

Without traffic rules:

- Cars would collide.
- Roads would become unsafe.
- People would lose trust.

Similarly, without AI Governance:

- AI may produce harmful outputs.
- Sensitive data could be exposed.
- Costs could spiral out of control.
- Regulatory violations could occur.

Simple Example

Imagine an employee asking:

"Show me all salaries in the company."

Without governance, an AI system might disclose confidential information.

With governance:

- Access is verified.

- Policies are enforced.
- Sensitive data is masked.
- Actions are audited.

Governance Pillars

Pillar	Description
Security	Protect enterprise data
Privacy	Ensure data confidentiality
Compliance	Meet regulatory requirements
Accountability	Define ownership
Transparency	Explain AI decisions
Monitoring	Track AI behavior

Intermediate Explanation

Enterprise AI Governance typically operates at three levels:

Level	Responsibility
Executive	Policies and approvals
Program	Governance implementation
Operational	Monitoring and compliance

Governance Framework

1. Define policies.
2. Establish approval processes.
3. Implement controls.
4. Monitor continuously.
5. Conduct audits.
6. Improve governance practices.

Governance Committee

Most enterprises establish an AI Governance Council consisting of:

- CIO
- CTO
- Legal Team
- Security Team
- Compliance Officers
- TPMs
- Product Leaders

Typical Governance Questions

- Who owns the AI product?
- What data is being used?
- Is the model compliant?
- What happens during failures?
- How are risks managed?

Why This Matters for an AI TPM

Responsibility	Impact
Delivery	Ensures AI products meet governance requirements before deployment.
Risk	Identifies and mitigates legal, security, and compliance risks.
Stakeholder Management	Coordinates governance reviews across multiple teams.
Budget	Prevents costly compliance failures and operational incidents.

Enterprise Example

Business Problem

A global banking organization wanted to deploy an AI assistant for employees but faced concerns regarding:

- Customer data privacy
- Regulatory compliance
- Auditability
- Access controls

Solution

Establish an enterprise AI Governance framework.

Architecture

- Azure OpenAI
- Azure AI Search
- Azure Key Vault
- Azure Monitor
- Azure Policy
- Azure Purview

Outcome

Metric	Result
Compliance Rate	100%
Audit Findings	Reduced by 75%
Security Incidents	Zero
Executive Confidence	High

Lessons Learned

- Governance should begin during ideation.
 - Governance committees accelerate decision-making.
 - Monitoring and audits are essential.
 - AI ownership must be clearly defined.
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Azure Implementation

Services Used

Service	Purpose
Azure OpenAI	Managed AI services
Azure AI Search	Controlled access to enterprise knowledge
Azure Functions	Workflow enforcement
Azure AI Foundry	Evaluations and governance capabilities

Supporting Services

Service	Purpose
Azure Policy	Policy enforcement
Azure Key Vault	Secrets management
Azure Monitor	Observability
Microsoft Purview	Data governance
Azure Active Directory	Identity management

Implementation Workflow

Step 1: Define Governance Policies

Examples:

- Data retention
 - Access control
 - AI usage guidelines
 - Incident response
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Step 2: Establish Governance Committee

Include:

- Legal
 - Security
 - Compliance
 - TPMs
 - Engineering
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Step 3: Implement Controls

Examples:

- RBAC
 - Private networking
 - Audit logging
 - Approval workflows
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Step 4: Conduct Reviews

Review:

- Risks
 - Compliance
 - Costs
 - Performance
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Step 5: Monitor Continuously

Track:

- Policy violations
- Security incidents
- Usage trends

- Audit findings
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Mermaid Diagram

```
```mermaid id="h5l2nu" graph TD
```

```
A[Business Requirement]
```

```
A --> B[AI Governance Council]
```

```
B --> C[Policies]
```

```
C --> D[Security Controls]
```

```
D --> E[Azure OpenAI]
```

```
E --> F[Monitoring]
```

```
F --> G[Compliance Reviews]
```

```

```

```
Governance Operating Model
```

```
```mermaid id="vf7gyj"
graph LR
```

```
A[Executive Team]
```

```
A --> B[AI Governance Board]
```

```
B --> C[TPMs]
```

```
C --> D[Engineering]
```

```
C --> E[Security]
```

```
C --> F[Legal]
```

```
D --> G[Production AI Systems]
```

```
E --> G
```

```
F --> G
```

TPM Checklist

- ☐ Define AI ownership.
 - ☐ Establish governance policies.
 - ☐ Conduct risk assessments.
 - ☐ Configure access controls.
 - ☐ Implement audit logging.
 - ☐ Review compliance requirements.
 - ☐ Enable monitoring.
 - ☐ Conduct governance reviews.
 - ☐ Document decisions.
 - ☐ Obtain approvals.
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Templates

Governance Review Checklist

Item	Status
Security Approved	Yes/No
Legal Approval Obtained	Yes/No
Compliance Reviewed	Yes/No
Monitoring Enabled	Yes/No
Ownership Defined	Yes/No

AI Governance Charter

Field	Description
Program Name	
Sponsor	
Governance Owner	
Risk Level	
Approval Date	

Risk Register

Risk	Impact	Mitigation
Data Leakage	High	RBAC and encryption
Compliance Failure	High	Governance reviews

Risk	Impact	Mitigation
Unauthorized Access	High	Identity controls
Cost Overruns	Medium	Budget monitoring

Common Mistakes

1. Treating governance as a post-deployment activity.
2. Failing to establish ownership.
3. Ignoring legal and compliance teams.
4. Neglecting audit requirements.
5. Underestimating security risks.
6. Skipping governance reviews.
7. Assuming AI vendors handle all compliance obligations.

Key Takeaways

- AI Governance is essential for building trustworthy enterprise AI systems.
- Governance should begin during the ideation phase.
- AI TPMs play a central role in coordinating governance activities.
- Security, privacy, and compliance are shared responsibilities.
- Azure provides multiple services to support governance and monitoring.
- Organizations with strong governance frameworks achieve faster and safer AI adoption.

Footer